

Liquid Cooling Designed for Maximum Uptime and Reliability

OCP Global Summit 2026
October 15, 2026

Company Headquarters

Nidec Development Headquarters

Key Applications:

- IT & Mobile Devices
- Home Appliances
- Robotics & Automation

Senior Vice President: Yuji Tanaka





Nidec Corporate Headquarters

- Founded: July 1973
- Headquartered: Kyoto, Japan

- World's leading comprehensive electric motor manufacturer (Small Precision Brushless DC Motors and HDD Spindle Motors)



Five Pillars

Next-Generation Energy Infrastructure

Contributing to infrastructure maintenance and environmental conservation



BESS(Battery Energy Storage System)



Alternator (Generator)

Turn-key generation systems (Generator)



Better Life

-Improving quality of life
-Powering reliability, safety and health













Home appliance motor







Refrigerator compressor

Base of AI Society

Contributing to society by responding to required evolution and exploding demand





Robotic motor



QD



HDD motors



In-Rack CDU





In-Row CDU

Mobility Innovation

Accelerating a low-carbon society through the electrification and automation of mobility





Traction motor



Anti-clouding fan



Seat fan







E-Bike motor

Advanced Manufacturing Solution

Strongly promoting high precision and labor-saving in manufacturing



Machine tools

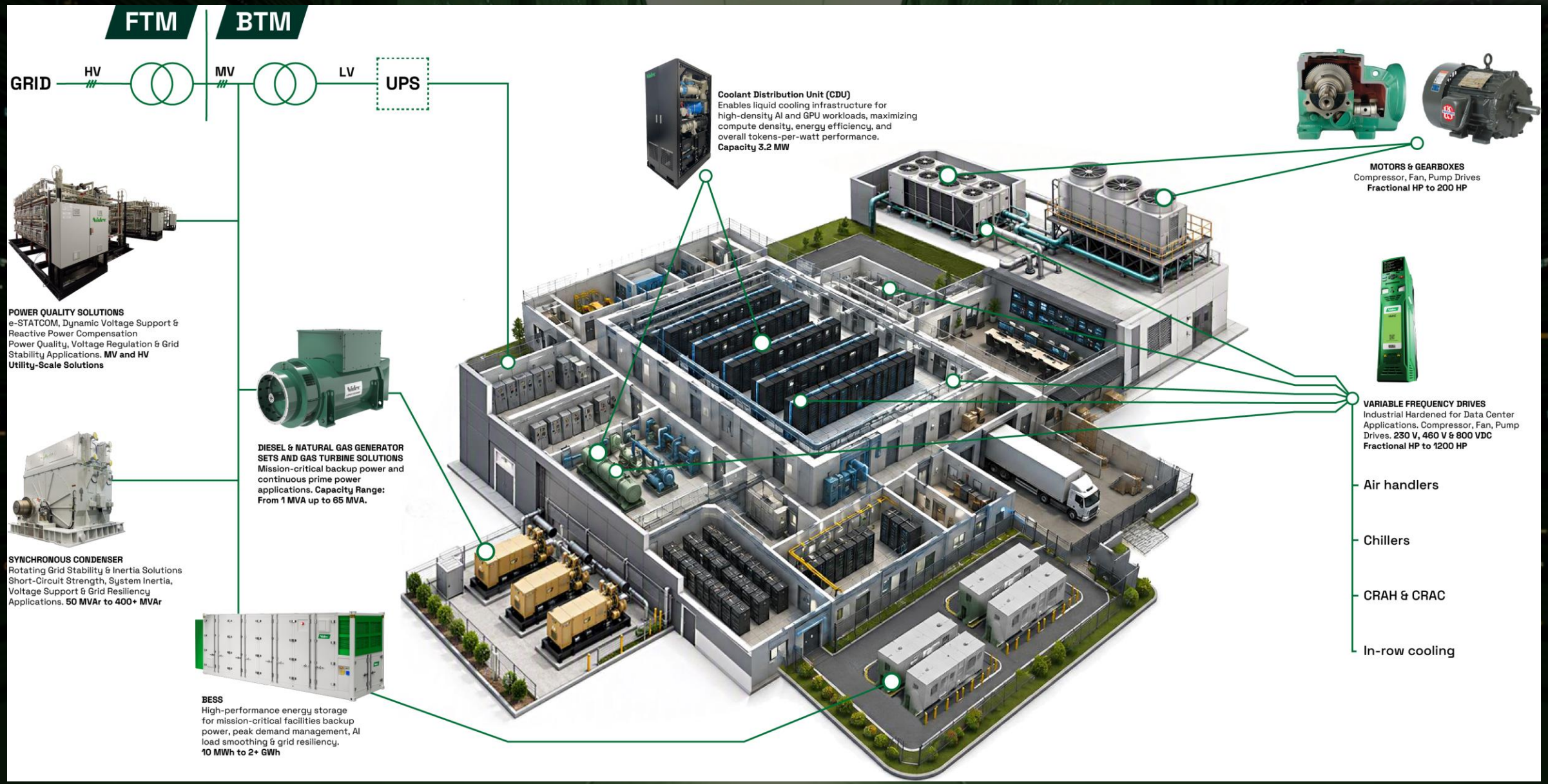




Reduction gear

Data Center Solutions Portfolio

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Thermal Specialist

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3.2MW

Heat Load/Unit

10W

In-Rack CDU





In-Row CDU

Cooling Fan

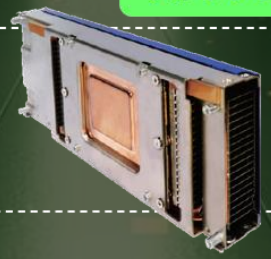


Blowers



HP/VC Modules







Thin HP/VC with Fan



Ultra Thin HP/VC

Air-Cooling

Liquid-Cooling

Nidec

Cooling Distribution Unit

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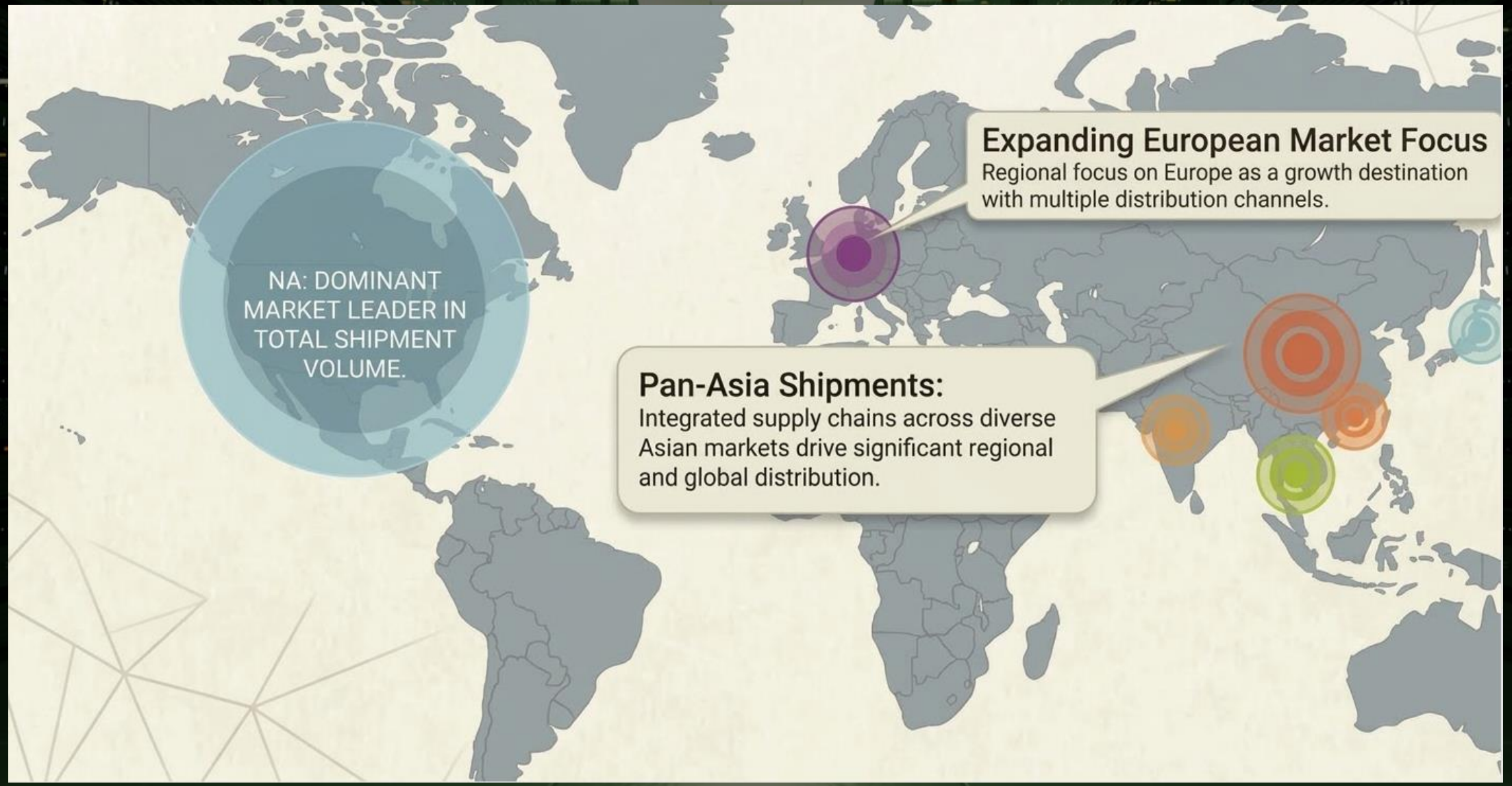




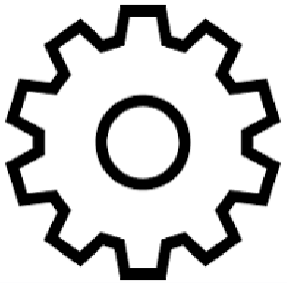


Cooling Distribution Unit

- Global Cumulative Shipment Record of >13,000 CDUs



After Sales Services - Support



Technical Support
Repair Services



24/7 Response

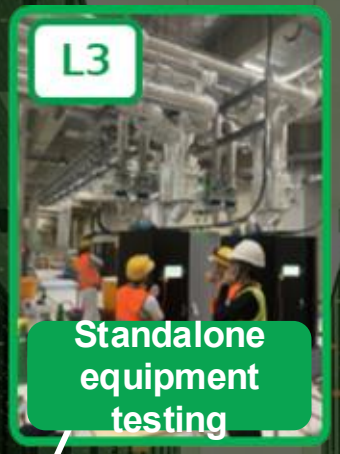


Regular Inspections
Scheduled Parts
Replacement

Installation and Commissioning Support











Integrated System Testing

Testing of critical MEP equipment, including the CDU.

Performance of functional testing items.

Witness acceptance testing at customer delivery site including installation in the customer environment

Factory Acceptance Testing (FAT) at Nidec's facility including stand-alone testing

Cooling Distribution Unit

Nidec

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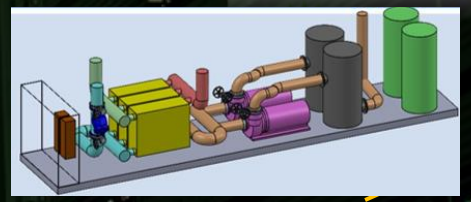
Rack Cooling Cap. [kW]

450
300
150
0











130kW

210kW

230kW

300kW

1,000kW
Nex Gen.

40kW
H100

In-Row CDU

In-Rack CDU

High Cooling cap.
In-Row CDU

In-Row CDU
(N+1)

New Tech.
CDU

In-Rack CDU
For Network, etc

C-CDU

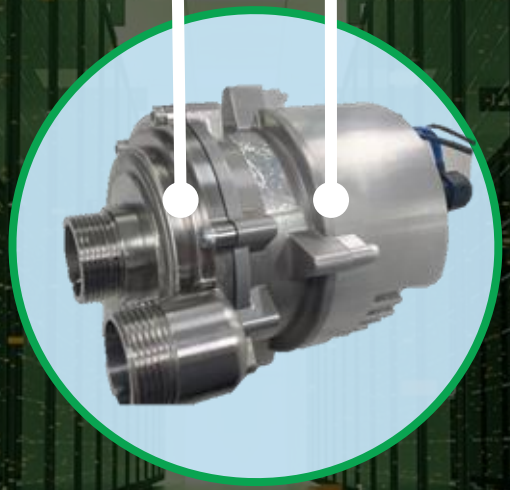
In-Rack CDU

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Generation 2.5



Target Spec	Gen. 2.5
Maximum Cooling Capacity	250 kW
Dimension	4U 19” W449 x D852 x H177
Communication	Live Monitoring, Redfish Capable
Pump	Redundant (1+1)
Redfish Support	Yes
Expansion Tank	Yes



Proprietary Compact High-Efficiency Pump

Generation 3.0



Target Spec	Gen. 3.0
Maximum Cooling Capacity	300 kW
Dimension	4U 19” W449 x D946 x H177
Communication	Live Monitoring, Redfish Capable
Pump	Redundant & Hot-Swappable
Redfish Support	Yes
Expansion Tank	Yes

In-Row CDU

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NIR2.5 2.0MW



-Compact
-High pressure
-Adaptable for Containerized DC

Deschutes 5.0 2.0MW

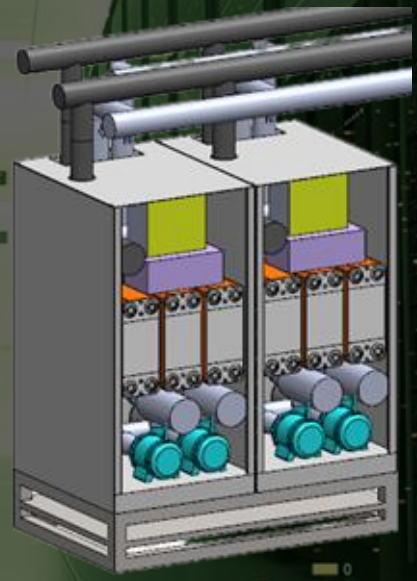


-High pressure
-Ultra low harmonic VFD

NIR3.0 3.2MW

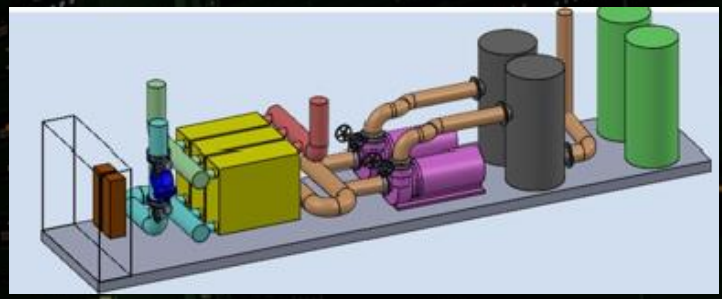


NIR3.0 S/D 6.4MW



- High cooling performance
- Skid typed CDU

NCCDU S/D 15MW



- Low Approach Temperature
- Data Center Hall Optimization
- Skid typed CDU

Nidec In-Row 3.0

Quality and Capacity:
Secure and High quality designed in Japan.

Massive 3.2MW Cooling Cap:
Delivers high-density cooling performance to support AI infrastructure.

Superior Thermal Efficiency:
*Achieves an approach temperature of **3°C at 3,200 kW***

100% Front-Access Maintenance:
All major components and sensors are designed for quick, hassle-free front access and replacement.



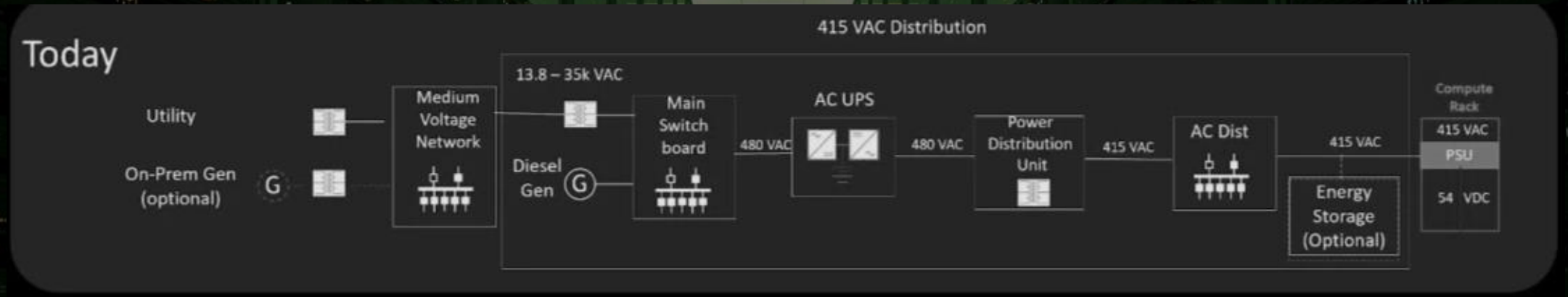
Future-Proof Power Compatibility:
*Under evaluation for **800 VDC** input option.*

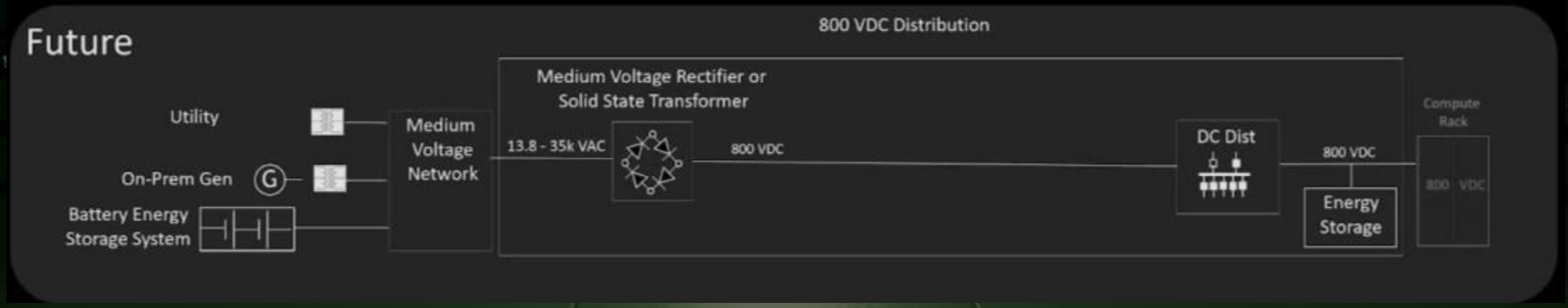
Clean Power Grid System:
Maintains THDi ≤5% to minimize total harmonic distortion and secure power supply stability.

Advanced Monitoring:
*Supports optional pH, conductivity, and turbidity sensors for **real-time coolant quality control.***

800VDC Power Distribution

- Nidec's future In-Row CDUs, including NIR 3.0, is planned for compatibility for both 415VAC and 800VDC configurations
- Nidec can leverage the existing design and dimensions with modifications to related electrical components as required





*Source: https://developer.nvidia.com/blog/building-the-800-vdc-ecosystem-forefficientscalable-ai-factories/

Quick Disconnect

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OCP UQD



XBMC



MQD



PBMC



Quick Disconnect

- Vertically integrated In-house machining capabilities converting raw material into finished goods using dedicated multi-functional machining capabilities by Nidec group company Takisawa
- Ultra precision machining and cleanliness requirements know-how from decades of HDD manufacturing
- With multiple production locations, Nidec can meet customer demands and various requirements as needed





Quick Disconnect

- Fully automated quick disconnect production line designed for contamination control, zero-defect inspection, and global resiliency
- Strict quality control with 100% leak testing of both fluid and gas particles
- Quality assurance with industry leading Class-100 cleanroom environment





STAGE 1: MASS PRODUCTION ENVIRONMENT

ISO Class 8 Control. Strict environmental blocking maintaining <3,520,000 particles (>0.5µm) per cubic meter. Mandatory clean suits and nitrile gloves.

STAGE 2: COMPONENT-LEVEL PURIFICATION

Multi-tank Ultrasonic + DI washing paired with dry ovens. Eradication of contamination, actively removing particles >0.3µm.

STAGE 3: IN-LINE ASSEMBLY WASHING

Dedicated QD inside-wash and dry equipment on the line. Continuous removal of microscopic particles generated by mechanical friction during assembly.


Nidec
All for dreams